

MONCHO UNSCHOOLING

Board Games

Unit Study

“Every board game is a tiny world with rules you can bend, break and rebuild, so what happens when YOU become the game designer?”

Math | History | Language Arts | Art & Creativity | Critical Thinking | Life Skills & SEL | Geography | Music | Movement & Body | Technology

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🧭 **How to use this guide:** This is not a workbook, and there's no answer key — on purpose! Every challenge is an invitation. Pick the ones that spark curiosity, skip the rest, and go find the answers by doing. **AI plans. Kids DO.** 🚀

🎲 MATH

Board games are really number machines, and you're about to take one apart to see how it ticks.

🎲 Challenge 1: Dice Detective

🎯 Probability and tallying

If a die has no memory, why does it sometimes feel like it's "hot" or "cold"? ❓

Grab a die, a pencil and a sheet of paper.

1. Draw six columns labeled 1 through 6.
2. Before you roll, guess how many times each number should come up in 60 rolls.
3. Roll the die 60 times and make a tally mark for each result.
4. Count your tallies. Which number showed up most? Which showed up least? How far off were your guesses?
5. Now hold the die and look at opposite faces. Add 1 and 6, then 2 and 5, then 3 and 4. What do you notice?
6. Write one sentence about your biggest surprise.

Why: each roll is independent, so a short run of rolls is usually lumpy, even though a fair die would even out over thousands of rolls.

🌱 **Make it Easier:** Roll only 30 times, and count the tallies by fives to make adding quick.

🚀 **Level it Up:** Roll 60 more times and add the new tallies to your old ones. Did the totals get closer to even, and by how much?

♟️ Challenge 2: Count the Chessboard

🎯 Multiplication

How can you find the number of squares on a chessboard without counting every one? ❓

A chessboard is a perfect grid, which makes it a multiplication playground.

1. Count the rows, then count the columns. Multiply them to find the total number of squares (a chessboard is 8 x 8).
2. Each player starts with 16 pieces: 1 king, 1 queen, 2 rooks, 2 bishops, 2 knights and 8 pawns. Add them up to check that they make 16.
3. Both players together have how many pieces? How many of those are pawns?
4. At the start, how many squares are left empty? Subtract to find out.
5. Write 3 multiplication story problems using the board, like "If every pawn took 3 steps..." Swap with someone and solve each other's.

🌱 **Make it Easier:** Skip the story problems and just find the total squares, the pieces for one player, and the empty squares.

🚀 **Level it Up:** Write a story problem that needs two steps, like multiplying and then subtracting, and give it to someone to solve.

🏙️ Challenge 3: Design Your City Board

🎯 Perimeter

If your board has a set number of spaces, how long is the path all the way around it? ❓

You're the city planner for a brand-new game town. The standard US Monopoly board has 40 spaces. Now it's your turn to plan one.

1. Grab a big sheet of paper, a ruler and a pencil.
2. Draw a square board. Each space is a 2 cm by 2 cm square, and the four corners are shared by two sides.
3. Decide how many spaces go on each side, counting the corners. Does your total come close to 40?
4. Measure the length of one side, then add all four sides to find the perimeter.
5. Name your spaces, color them in, and label your measurements right on the board.

🌱 **Make it Easier:** Use 3 cm squares and put 6 spaces on each side, so the adding is simple.

🚀 **Level it Up:** Change the shape to a rectangle with a longer top and bottom, and work out the new perimeter and total number of spaces.

📜 HISTORY

Long before video games, people carved boards, rolled dice and played tiny worlds of rules that still survive today.

📜 Challenge 1: Play a 4,500-Year-Old Game

🎯 Archaeology

How can a game stay playable after thousands of years under the ground? ❓

You're the archaeologist's apprentice. The Royal Game of Ur was played in ancient Mesopotamia about 4,500 years ago. Archaeologist Leonard Woolley found boards in the Royal Cemetery of Ur in the 1920s, and the British Museum holds a famous one.

1. Search "British Museum Royal Game of Ur" with a grown-up and read the rules they explain.
2. Cut a cardboard strip and draw a board: a block of 4x3 squares, a narrow bridge of 2 squares, and a block of 2x3 squares at the end.
3. Make dice: color one tip of each of 4 craft sticks or pyramid-shaped paper dice. Count how many colored tips land up.
4. Make 7 pieces for each player from bottle caps or coins, and decorate the squares like the museum board.
5. Play one round, following the rules you read.

Why: the museum board was only a copy of a game people already knew. Its rules were later read from a clay tablet, so the game could be played again.

🌱 **Make it Easier:** Skip the bridge and draw a simple path of 12 squares. Use a normal die and race one piece each.

🚀 **Level it Up:** Write your own "tablet" of rules in your best handwriting, then teach a friend using only that page.

📜 Challenge 2: Pack a Game for the Tomb

🎯 Primary Sources

Why would anyone want to take a board game with them into the next world? ❓

You're the Egyptian game keeper, and you must prepare a game to be buried in a tomb. Senet is an ancient Egyptian board game with 30 squares in 3 rows of 10, and a Senet set was found in the tomb of Tutankhamun.

1. Draw 3 rows of 10 squares on cardboard to make your Senet board.
2. Choose 5 pieces for each player from pebbles, buttons or clay.
3. Pick 3 more things you'd place beside the game to help the owner play, such as a cup, a pebble or a sign.
4. Decide where each item goes in your tomb and why, using a shoebox as the tomb.
5. Invite a friend to walk through your tomb. Explain each item as if you were a museum guide.

🌱 **Make it Easier:** Choose only 3 items in total and tell your friend one reason for each.

🚀 **Level it Up:** Look up one real object found in Tutankhamun's tomb and explain how it fits in the same story as your game.

📜 Challenge 3: From Chaturanga to Chess

🎯 Mapping History

How can a game travel across the world without a single airplane? ❓

Chess developed from an Indian game called chaturanga, which was played about 1,500 years ago.

1. Lay a rope on the floor from one end of the room to the other. It is your route.
2. Make 5 index cards: India, Persia, the Arab world, Europe and the modern world.
3. Look up with a grown-up how the game changed in each place, such as its name and the pieces.
4. Write one change on each card and put the cards along the rope in order.
5. Walk the route slowly, telling the story out loud at each stop, and move a chess piece along the rope as you go.

🌱 **Make it Easier:** Use only 3 cards: India, Persia and Europe. Name one change at each stop.

🚀 **Level it Up:** Add a card for another place the game reached, and tell what the people there might have changed to make it their own.

📖 LANGUAGE ARTS

Every board game is a tiny world built out of words, from the names on its tiles to the rules that hold it together.

🔤 Challenge 1: The Word Log

🎯 Vocabulary building

Can the longest word you build also turn out to be the best one?❓

Scrabble was created by architect Alfred Mosher Butts, and the standard game has 100 letter tiles on a 15 x 15 board. Play a round with a real set, or cut 100 small squares of paper and write letters on them, with more copies of common letters like E and A and just one of rare ones like Q and Z.

1. Draw 7 tiles and build words on a table or a drawn grid, taking turns with a family member.
2. Keep a log with three columns: longest word, best word (you decide what "best" means) and strangest word.
3. After the round, look up any word you aren't sure about in a dictionary.
4. For every word in your log, write what it means in your own words.

🌱 **Make it Easier:** Draw only 5 tiles at a time and log just one longest word and one favorite word.

🚀 **Level it Up:** Add a column for each word's part of speech, and find one word in your log you could swap for a more exact synonym.

📜 Challenge 2: Write the Rule Book

🎯 Clear writing

How can words alone teach someone a game they have never seen? ❓

You're the rule-book author. Pick a brand-new game, either one you invent or a simple one you know well, like a race across a drawn path with a die and a few pawns.

1. List what players need: pieces, board, number of players.
2. Write what each player does on a turn, in short numbered steps.
3. Write how the game ends and how someone wins.
4. Give your rules to a friend who has never seen the game, and stay silent while they try to play.
5. Every time they get stuck or ask a question, fix that sentence.

Why: your reader can't see inside your head, so any rule you skipped leaves a gap that confuses them.

🌱 **Make it Easier:** Pick a very simple game and write just 5 short rules, adding a quick drawing of the board.

🚀 **Level it Up:** Add a "Tricky Situations" section that covers 3 surprises, like a tie or a player who can't move.

🕵️ Challenge 3: Name Detectives

🎯 Word origins

Where did the strange names of famous games come from? ❓

Go is a strategy game from China (called weiqi there) that is more than 2,500 years old; it is played on a 19 x 19 grid with black and white stones. Become a name detective and find out what game names really mean.

1. Look up the origin of "Go" in a dictionary or a library book on word histories.
2. Investigate "checkmate", which comes from the Persian "shah mat", and find 2 more game names with a surprising past.
3. Ask an adult or older relative which game names they know and what they think those names mean.
4. Write one clue-card per name with its language of origin and its meaning.
5. Practice, then give a 3-minute talk to your family about what you discovered.

🌱 **Make it Easier:** Investigate only "Go" and "checkmate", with an adult helping you read.

🚀 **Level it Up:** Draw a family tree of one name's journey from its first language to English, and guess how its meaning changed along the way.

🎨 ART & CREATIVITY

Every board game is a tiny world, and the pieces, boards and boxes are all waiting for your imagination to design them.

🗿 Challenge 1: Pieces With Personality

🎯 Sculpting

Could you pick out a knight from a bishop with your eyes closed? ❓

Your mission is to sculpt a full chess-style set where every type of piece feels different in your hand. Pick a type for each personality: a tall, calm king, a swift knight, a sturdy rook, and so on.

1. Make salt dough (1 cup flour, ½ cup salt, ½ cup water) or grab air-dry clay.
2. Decide on 5-6 piece types, and give each one a different height and a different shape that sticks out, like a pointed hat, a flat top or a curved neck.
3. Sculpt two of each type, making them similar enough to look like a team but with one small mark that sets your team apart from the other (a dot, a line, a tiny crown).
4. Let them dry, then close your eyes and have a friend hand you pieces one at a time. Can you name each one by touch alone?
5. Paint your two teams in colors that are easy to tell apart.

Why: your fingertips notice bumps, edges and height differences, so strong shapes make pieces easy to recognize without looking.

🌱 **Make it Easier:** Make only 3 piece types, with one piece of each, and use big simple shapes like a ball, a cone and a block.

🚀 **Level it Up:** Add a sculpted "special move" clue to each piece, like wings on a piece that can jump, and design a base that shows how that piece moves.

🌀 Challenge 2: Spiral of Surprises

🎯 Illustration

How do you make a path that twists toward the middle feel like a wild adventure?❓

The Game of the Goose is a spiral race game from Europe dating back to the late 1500s. Now it's time to draw your own version.

1. On a big sheet of paper, draw a spiral path that curls toward the center, then divide it into 40 or more squares.
2. Draw a different picture in each square, like a dragon, a lighthouse, a bridge or a treasure chest.
3. Choose 6-8 squares for surprises, like one that sends you back 3 spaces, one that lets you roll again and one that makes you skip a turn.
4. Make the center square the finish line, and draw it so it feels like a big reward.
5. Color it in, then play a round with a die and two small objects as markers.

🌱 **Make it Easier:** Draw just 20 squares and give only 3 surprise squares.

🚀 **Level it Up:** Give your spiral a theme, like the deep sea or outer space, and write a short rule for each surprise that fits the story.

📦 Challenge 3: Open Me!

🎯 Graphic Design

What makes you reach for one box on a shelf and walk right past another?❓

You're the packaging designer, and a game shop wants a cover that jumps off the shelf. Use a game you've invented, or invent one now.

1. Come up with a catchy title and sketch three different logo ideas for it.
2. Pick your best logo and draw the front cover on cardboard or thick paper, using bold colors and one big, exciting picture.
3. On the back, write a short blurb of 3-4 sentences. Start with a hook, say what players do, and end with a challenge like "Will you reach the end first?"
4. Add the details every box needs: the number of players, the ages and how long a game takes.
5. Show it to someone and ask: "Would you open this box?"

Why: people decide quickly, so big pictures, short words and strong colors grab attention in seconds.

🌱 **Make it Easier:** Design only the front cover with a title and one picture.

🚀 **Level it Up:** Make a full box by folding cardboard into a real box shape, and add a side panel with a slogan and small icons.

🧠 CRITICAL THINKING

Every board game is a puzzle waiting to be solved, and thinking like a detective, a judge and a debater will make you a much stronger player.

🎯 Challenge 1: The Never-Lose Strategy

🎯 Strategic Thinking

Is there a way to play so well that your opponent can never beat you? ❓

Tic-tac-toe-like games were played in ancient Rome under the name "terni lapilli," with three pieces each moving on a grid. Before you try the Roman version, master the one you know.

1. Draw a 3x3 grid and play 10 games of Tic-tac-toe against a family member. Switch who goes first every game.
2. After each game, jot down what move helped you most and what mistake cost you.
3. Look for a pattern: which first moves never lose? How do you block a trap?
4. Now try the Roman version on the same 3x3 grid. Each player gets only 3 pieces (use coins or buttons) and, once all 6 are placed, must slide one piece per turn to a neighboring empty square. Why is it trickier?
5. Write your best "never lose" tips on an index card.

Why: In Tic-tac-toe the board fills up and games end in draws, but when pieces must keep moving, positions keep changing and you have to think a few moves ahead.

🌱 **Make it Easier:** Play only the regular Tic-tac-toe and find just one rule that stops you from losing.

🚀 **Level it Up:** Invent your own version with 4 pieces each on a 4x4 grid, then test whether a never-lose strategy still exists.

⚔️ Challenge 2: The Fairness Trial

🎯 Evaluating Fairness

If one side starts with more warriors, can a game still be fair? ❓

You're the game judge. The Hnefatafl family of "Viking chess" games was played across northern Europe, with a king in the center defended by his warriors against attackers.

1. Draw a 9x9 grid on paper (the Tablut version), or a smaller 7x7 one. Use coins or buttons as pieces, with one special piece for the king. Put the king in the middle with 8 defenders around him, and place 16 attackers around the edges.
2. Pieces move like rooks in chess, any number of squares in a straight line. Attackers win by surrounding the king, and the king wins by reaching a corner.
3. Play twice with a family member, swapping sides between games.
4. Debate: which side had it easier? Which side had more fun? Give each side one argument.
5. Announce your verdict to the family and suggest one rule change to make it fairer.

🌱 **Make it Easier:** Play on a 7x7 board with fewer pieces, and judge only which side felt easier.

🚀 **Level it Up:** Test your rule change in a new game and decide whether it made the game more balanced.

🤖 Challenge 3: When the Machine Always Wins

🎯 Forming Opinions

Would you still play a game if you knew a machine could never lose? ❓

In 1997, the IBM computer Deep Blue defeated world chess champion Garry Kasparov in a match. It was the first time a computer beat a reigning world champion under match conditions.

1. Look at a picture of a chessboard and read this story together, or ask a grown-up to explain it.
2. Think about your favorite game. If a computer could beat everyone at it, would you still want to play?
3. Write down one reason a game can still be fun even when a machine is better than you, and one reason it might not be.
4. Hold a short family opinion round where everyone shares one reason, and no one may interrupt.

🌱 **Make it Easier:** Just answer one question aloud: "Would I still play?" and say why.

🚀 **Level it Up:** Hold the debate with two teams, one defending "yes, it's still fun" and one defending "no, it isn't," then switch sides and argue the opposite.

🤝 LIFE SKILLS & SEL

Every game ends with a winner and a loser, which makes board games the perfect place to practice staying kind, calm and brave.

🏆 Challenge 1: Lose Like a Legend

🎯 Handling Disappointment

Can you stay cool and kind when the game doesn't go your way? ❓

Pick a quick game that has a clear winner, like a card game, tic-tac-toe or a short board game, and gather one or two players.

1. Before you start, brainstorm 3 different ways to lose gracefully. Try a handshake and "Good game!", naming one thing the winner did well, or taking a deep breath and smiling.
2. Act each one out once, out loud, like a tiny rehearsal on stage.
3. Play the game.
4. When it ends, use one of your 3 ways, even if you win. If you win, use it to cheer up the other player.
5. On a small card, write the phrases you actually used and sign it.

Why: rehearsing a response before big feelings show up gives your brain a ready-made plan, so you don't have to invent one while you feel upset.

🌱 **Make it Easier:** Practice just 1 way to lose gracefully, like saying "Good game!", and skip the card.

🚀 **Level it Up:** Play three rounds and try a different graceful move each time. Ask a friend or sibling to guess which one you picked.

🎲 Challenge 2: Pachisi Patience Practice

🎯 Self-Control

What can you do in the few seconds before a dice roll to keep your cool? ❓

Pachisi is a traditional Indian race game played on a cross-shaped board, and it inspired Ludo and the Parcheesi board game.

1. Draw a big cross-shaped Pachisi-style board on a bedsheet or cardboard, or chalk it outside. Add a path of squares along each arm.
2. Use buttons or small stones as pieces and play a race round with your family. You will need dice, or you can make a number cube from paper.
3. Before every roll, the roller says out loud one calm-down trick they will use if the dice go badly: a deep belly breath, counting to five, or a big stretch.
4. After the round, talk about which tricks really helped.
5. Make a mini card with the three tricks that worked best and stick it where your family plays.

Why: slow breathing tells your body it is safe, so your heartbeat calms down and frustration shrinks.

🌱 **Make it Easier:** Use only one trick, like counting to five, and play a short race with just two pieces each.

🚀 **Level it Up:** Invent a brand-new calm-down trick, teach it to everyone, and vote on whether it earns a spot on the card.

🎉 Challenge 3: Game Night Host

🎯 Leadership and Inclusion

What does a great host do so that every single person has fun?❓

Tonight, you are in charge of the family game night.

1. Interview each person who will play. Ask what games they like, what they find hard, and whether they want to play long or short games.
2. Use their answers to choose games everyone can play. Mix it up if people have different ages or skills.
3. Explain the rules clearly in your own words, and check that everyone understands before you start.
4. During the game, keep an eye out for anyone feeling left out, and invite them in by asking for their ideas or giving them a turn.
5. At the end, ask everyone for a thumbs up, thumbs sideways or thumbs down on how the night went, and write down one thing to do better next time.

Why: when people feel heard before an activity starts, they join in with much more energy and kindness.

🌱 **Make it Easier:** Choose just one game and ask each person one question about it before you start.

🚀 **Level it Up:** Plan a whole evening with two games and a snack break, and write a short schedule everyone can follow.

🌍 GEOGRAPHY

Every board game was born somewhere, and following where each one came from lets you travel the whole world without leaving your table.

🪨 Challenge 1: Dig, Sow and Capture!

🎯 Map Reading

Why would people on two different continents invent such similar games? ❓

Mancala is a family of "sowing" games played across Africa and Asia for many centuries, often using seeds, pebbles or holes dug in the ground. Time to play it the old way.

1. Make two rows of 6 holes in a sand tray, a muffin tin or an egg carton. Each hole starts with 4 pebbles or dried beans.
2. Take turns scooping up all the seeds from one hole on your side and dropping one into each hole as you move around the board.
3. If your last seed lands in an empty hole on your side, capture the seeds in the hole opposite. The player with the most captured seeds wins.
4. Look up the Mancala family in an atlas or online. Put a dot on a blank world map for each region where a version is played, and label the dots in your own handwriting.

🌱 **Make it Easier:** Play with only 3 seeds per hole and just find Africa and Asia on the map.

🚀 **Level it Up:** Find two different Mancala names from different countries and mark each one on your map.

🧶 Challenge 2: Game Journey Map

🎯 Scale and Distance

Which game had the longest trip to reach you? ❓

Many games traveled huge distances. Get a world map, a printed one, an atlas, or Google Earth, which is free to open.

1. Pick four games: Pachisi from India, Backgammon from the Middle East, Nine Men's Morris from Europe, and Mancala from Africa and Asia.
2. Mark where each one started with a sticker or colored pin.
3. Stretch yarn from each pin to your home, showing a possible path a game could travel.
4. Hold each piece of yarn against the map's scale bar to turn its length into kilometers or miles.
5. Write each distance next to its pin and circle the longest journey.

Why: a flat map stretches the world, so a short line on paper can hide a very long real trip.

🌱 **Make it Easier:** Measure with a ruler and just say which yarn looks longest and which looks shortest.

🚀 **Level it Up:** Add a path for each game that follows coastlines and rivers, then compare it to the straight yarn line.

🏪 Challenge 3: The Catan Trade Merchant

🎯 Resources and Regions

If you wanted real wood, brick, sheep, wheat and ore, where in the world would you shop? ❓

You're the trade merchant for a new island. Catan (originally "Die Siedler von Catan") was designed by Klaus Teuber and first published in Germany in 1995, and players collect resources like wood, brick, sheep, wheat and ore.

1. Research one real country or region that produces each resource. Use an atlas or a library book on world products.
2. Write each resource on one card and the matching country on another.
3. Use the map scale to estimate how many kilometers each resource would travel to reach you. Add the five distances to find your merchant's total trip.
4. Make a little answer card for each pair so a friend can check their matches.

🌱 **Make it Easier:** Match only three resources and skip the distances.

🚀 **Level it Up:** Find a second country for each resource, then choose which one is the closest supplier for you.

🎵 MUSIC

Every board game has its own soundtrack of rattles, clicks and cheers, just waiting for you to turn it into music.

🥁 Challenge 1: The Sound of Game Night

🎯 Rhythm Composition

What kind of band could you form using only the noises hiding inside a game box? ❓

Now you'll turn those game sounds into rhythm.

1. Grab a game box and test every sound: shake dice in a cup, clack tiles together, tap a pawn on the table, slide cards, rattle the empty box.
2. Pick your 4 favorite sounds and give each a name (like "rattle" or "click").
3. Write a 4-beat pattern using those names, like: rattle, click, click, slide.
4. Practice it slowly, then repeat it 4 times in a row without stopping.
5. Gather your family and perform it live. Everyone can take one sound and play it together!

🌱 **Make it Easier:** Use just 2 sounds and a pattern of 4 beats, like click, click, rattle, rest.

🚀 **Level it Up:** Layer 3 patterns at once, with one player keeping a steady beat while others play different rhythms on top, and add one beat of silence on purpose.

🎶 Challenge 2: The Victory Chant

🎯 Songwriting

How do you make a song so catchy that everyone wants to sing it when someone wins?❓

You're the victory bard, and every game needs your music to crown its winner!

1. Pick a game your family loves to play.
2. Write 4 short lines about winning. The 2nd and 4th lines should rhyme, like "We rolled, we moved, we played it right, now celebrate this winning night!"
3. Choose a beat to go with it: clap, stomp, or tap on the table.
4. Sing it out loud until it feels smooth, then change any words that feel clunky.
5. Teach it to your family, line by line, until everyone can sing it together.

🌱 **Make it Easier:** Write just 2 lines and use a clap on every word.

🚀 **Level it Up:** Add a call-and-response, where you sing a line and everyone else answers, and make up one victory dance move to go with it.

🏃 MOVEMENT & BODY

When you are the game piece, every hop, tiptoe and crab-walk becomes part of the rules you can bend, break and rebuild.

🐍 Challenge 1: Be the Piece on a Giant Snakes and Ladders Board

🎯 Gross Motor Coordination

What does a game feel like when your whole body is the piece instead of a tiny token? ❓

Long ago in India, a game called Moksha Patam used ladders for good deeds and snakes for bad deeds. Now you'll build it big enough to walk on.

1. Use chalk or masking tape to draw a grid of 5 squares by 5 squares on the floor or driveway, with each square big enough to stand in.
2. Number the squares in a zigzag, starting at the bottom left. Write 1 to 5 going right, then 6 to 10 going left, and keep switching.
3. Draw 3 ladders and 3 snakes joining squares, and choose a different movement for each, such as a bunny hop for ladders and a slow-motion slither for snakes.
4. Take turns. Toss a coin to move: heads = 2 squares, tails = 1 square. Move by hopping, dancing or tiptoeing.
5. Land on a ladder foot and you move up. Land on a snake head and you slide down. First to square 25 wins.

🌱 **Make it Easier:** Draw a smaller 4-by-4 board with only 2 ladders and 2 snakes, and just walk between squares.

🚀 **Level it Up:** Give each ladder a "good deed" you must act out, like helping someone, and each snake a "bad deed" you must act out the opposite of, just like the ancient game did.

🔴 Challenge 2: Human Nine Men's Morris

🎯 Balance and Quick Thinking

How hard is it to plan a clever move when your body has to hop, tiptoe or crab-walk to get there? ❓

Boards for Nine Men's Morris have been found carved into stone in medieval churches and cloisters across Europe, so people have played it for a very long time.

1. Chalk three squares inside each other on the driveway, or lay them out with rope in the yard. Join the middle point of each side with a line, and mark a dot on every corner and every spot where lines cross.
2. Two players take turns standing on a dot. Each player tries to get three of their own pieces in a straight line, called a mill.
3. Choose a movement rule before each round. Players must hop, tiptoe or crab-walk to their next dot, and a wobble counts as a lost turn.
4. Keep a list of the silliest movement rules you invent, like "walk backwards" or "freeze like a statue for 3 seconds."

Why: Balancing and planning use the same brain, so your thinking can feel wobblier when your body is working hard too.

🌱 **Make it Easier:** Play with a toy or stone as the piece and only use one movement rule, such as hopping.

🚀 **Level it Up:** Add a time limit of 10 seconds per move, or invent a new rule for forming a mill and teach it to a friend.

💻 TECHNOLOGY

Every board game has hidden machinery inside it, and with a spreadsheet or a chess program you can open the box and see how it really works.

🎲 Challenge 1: Dice Fairness Lab with a Spreadsheet

🎯 Data Charts

If you roll a die 60 times, will every number get the same number of turns to shine? ❓

You're testing whether a die plays fair, using a real data tool.

1. Use your 60 tallies from the Dice Detective challenge in Math. If you skipped it, roll a die 60 times and tally each result.
2. Open a free spreadsheet (Google Sheets or LibreOffice Calc). In column A, type the numbers 1 to 6. In column B, type how many times each one came up.
3. Select both columns and choose Insert, then Chart. Pick a bar chart.
4. Find a friend or family member with a different die and compare charts. Does any face seem to win more often? Write one sentence about what you found.

Why: a single die can't "remember" its last roll, so small streaks are normal. Over many rolls the bars tend to even out, but they are almost never perfectly equal.

🌱 **Make it Easier:** Roll only 30 times, and draw your bar chart on graph paper by coloring one square for each roll instead of using the spreadsheet.

🚀 **Level it Up:** Add a column that turns each count into a percentage (count ÷ 60 × 100). Then roll 60 more times and see whether your chart gets flatter or lumpier.

♟️ Challenge 2: Chess vs Computer Detective

🎯 Analyzing Behavior

When a computer gets "smarter," what exactly does it do differently on the board? ❓

You're a detective, and your clue is how the computer plays.

1. Go to lichess.org with a grown-up and start a game against the computer at its easiest level. Play 10 to 15 moves.
2. Start a new game at a harder level. Write down what changed: Does it answer faster? Make fewer silly mistakes? Set traps?
3. Open the "analysis board" after a game. Scroll through your moves and find one big turning point where the evaluation bar jumps.
4. Think about how you'd explain to a younger kid how a machine can beat even expert players. Draw it as a sketch or record a short voice message.

Why: the computer checks thousands of possible moves ahead very fast and scores each position. At harder levels, it looks deeper and makes fewer mistakes.

🌱 **Make it Easier:** Play only at the easiest level and just notice one thing the computer did that surprised you.

🚀 **Level it Up:** At the turning point, pause and guess the best move before the analysis shows it. Then try that move in a new game and see if it works.